Chapter 14- Multiple Integrals

Exercise 14.5: Triple Integrals

Question 1:

$$\int_{-1}^{1}\int_{0}^{2}\int_{0}^{1}(x^2+y^2+z^2)\,dx\,dy\,dz$$

Solution:

$$\int_{-1}^{1}\int_{0}^{2}\int_{0}^{1}(x^2+y^2+z^2)dxdydz$$

$$=\int_{-1}^{1}\int_{0}^{2}\left[\int_{0}^{1}(x^2+y^2+z^2)dx\right]dydz$$

Solve the inner integral, integrate with respect to x, and treat y and z as constants

$$\int_{0}^{1}(x^2+y^2+z^2)dx=\left[\frac{x^3}{3}+xy^2+xz^2\right]_0^1$$

Evaluate the limits in the variable x

$$=\left(\frac{(1)^3}{3}+(1)\,y^2+(1)\,z^2\right)-\left(\frac{(0)^3}{3}+(0)\,y^2+(0)\,z^2\right)$$

$$=\frac{1}{3}+y^2+z^2$$

Then,

$$=\int_{-1}^{1}\int_{0}^{2}\left(\frac{1}{3}+y^2+z^2\right)dydz$$

$$=\int_{-1}^{1}\left[\int_{0}^{2}\left(\frac{1}{3}+y^2+z^2\right)dy\right]dz$$

Integrate with respect to y, treat z as a constant

$$\int_0^2 \left(\frac{1}{3}+y^2+z^2\right)dy = \left(\frac{1}{3}y+\frac{y^3}{3}+z^2y\right)_0^2$$

$$= \left(\frac{1}{3}(2)+\frac{(2)^3}{3}+z^2\,(2)\right) - \left(\frac{1}{3}(0)+\frac{(0)^3}{3}+z^2\,(0)\right)$$

$$= \frac{2}{3}+\frac{8}{3}+2z^2$$

$$= \frac{10}{3}+2z^2$$

$$\int_{-1}^1 \left[\int_0^2 \left(\frac{1}{3}+y^2+z^2\right)dy\right]dz = \int_{-1}^1 \left(\frac{10}{3}+2z^2\right)dz$$

Integrate

$$= \left(\frac{10}{3}z+\frac{2z^3}{3}\right)_{-1}^1$$

Evaluate

$$= \left(\frac{10}{3}(1)+\frac{2(1)^3}{3}\right) - \left(\frac{10}{3}(-1)+\frac{2(-1)^3}{3}\right)$$

$$= 4-(-4)$$

$$= 8$$

Question 2:

$$\int_{1/3}^{1/2} \int_0^{\pi} \int_0^1 zx \sin xy \, dz \, dy \, dx$$

Solution:

$$= \int_{1/3}^{\frac{1}{2}} \int_0^{\pi} \int_0^1 zx \sin xy \;\; dz \, dydx$$

$$= \int_{1/3}^{\frac{1}{2}} \int_0^{\pi} |\frac{z^2}{2} x \sin xy|_0^1 \, dydx$$

$$= \int_{1/3}^{\frac{1}{2}} \int_0^{\pi} \frac{1}{2} x \sin xy \, dydx$$

$$= \int_{1/3}^{\frac{1}{2}} \frac{1}{2} (-\cos xy)_0^{\pi} dx$$

$$= \int_{1/3}^{\frac{1}{2}} \frac{1}{2} (-\cos \pi x - (-1)) dx$$

$$= \int_{1/3}^{\frac{1}{2}} \frac{1}{2} (1 - \cos \pi x) dx = \int_{\frac{1}{3}}^{\frac{1}{2}} \frac{1}{2} dx - \frac{1}{2} \int_{\frac{1}{3}}^{\frac{1}{2}} \cos \pi x) dx$$

$$= \int_{\frac{1}{3}}^{\frac{1}{2}} \frac{1}{2} dx - \frac{1}{2} \int_{\frac{1}{3}}^{\frac{1}{2}} \cos \pi x dx$$

$$= \frac{x}{2} |_{\frac{1}{3}}^{\frac{1}{2}} - \frac{\frac{1}{2} (\sin \pi x)}{\pi} {}_{\frac{1}{3}}^{\frac{1}{2}}$$

$$= \left(\frac{1}{4} - \frac{1}{6}\right) - \left(\frac{1}{2\pi} \sin \left(\frac{\pi}{2}\right) - \frac{1}{2\pi} \sin \frac{\pi}{3}\right)$$

$$= \frac{1}{12} - \left(\frac{1}{2\pi} - \frac{1}{2\pi} \frac{\sqrt{3}}{2}\right) = \frac{1}{12} - \frac{2 - \sqrt{3}}{4\pi} = \frac{1}{12} + \frac{\sqrt{3} - 2}{4\pi}$$

Question 3:

$$\int_0^2 \int_{-1}^{y^2} \int_{-1}^{z} yz \, dx \, dz \, dy$$

Solution:

$$= \int_0^2 \int_{-1}^{y^2} \int_{-1}^{z} yz \, dxdzdy$$

$$= \int_0^2 \int_{-1}^{y^2} (yz^2 + yz) \, dz \, dy$$

$$= \int_0^2 \left(\frac{1}{3}y^7 + \frac{1}{2}y^5 - \frac{1}{6}y\right) dy$$

$$= \frac{47}{3}$$

Question 4:

$$\int_0^{\pi/4}\int_0^1\int_0^{x^2} x\cos y\,dz\,dx\,dy$$

Solution:

$$\int_0^{\pi/4}\int_0^1\int_0^{x^2} x\cos y dz dx dy$$

$$=\int_0^{\pi/4}\int_0^1\left[\int_0^{x^2} x\cos y dz\right] dx dy$$

Solve the inner integral, integrate with respect to z,
treat y and x as constants

$$\int_0^{x^2} x\cos y dz=[xz\cos y]_0^{x^2}$$

Evaluate the limits in the variable z

$$=x\left(x^2\right)\cos y-x(0)\cos y$$

$$=x^3\cos y$$

Then,

$$\int_0^{\pi/4}\int_0^1\left[\int_0^{x^2} x\cos y dz\right] dx dy=\int_0^{\pi/4}\int_0^1 x^3\cos y dx dy$$

$$=\int_0^{\pi/4}\left(\int_0^1 x^3\cos y dx\right) dy$$

Integrate with respect to x, treat y as a constant

$$\int_0^1 x^3 \cos y dx = \left(\frac{x^4}{4}\cos y\right)_0^1$$

$$= \frac{(1)^4}{4}\cos y - \frac{(0)^4}{4}\cos y$$

$$= \frac{1}{4}\cos y$$

$$\int_0^{\pi/4}\left(\int_0^1 x^3 \cos y dx\right)dy = \frac{1}{4}\int_0^{\pi/4} \cos y dy$$

Integrate

$$= \frac{1}{4}(\sin y)_0^{\pi/4}$$

Evaluate

$$= \frac{1}{4}\left(\sin\frac{\pi}{4} - \sin 0\right)$$

$$= \frac{1}{4}\left(\frac{\sqrt{2}}{2}\right)$$

$$= \frac{\sqrt{2}}{8}$$

Question 5:

$$\int_0^3 \int_0^{\sqrt{9-z^2}} \int_0^x xy\,dy\,dx\,dz$$

Solution:

$$\int_0^3 \int_0^{\sqrt{9-z^2}} \int_0^x xydydxdz$$

$$= \int_0^3 \int_0^{\sqrt{9-z^2}} \left[\int_0^x xydy\right] dxdz$$

Solve the inner integral, integrate with respect to y,
treat z and x as constants

$$\int_0^x xydy = \left[\frac{xy^2}{2}\right]_0^x$$

Evaluate the limits in the variable y

$$= \frac{x(x)^2}{2} - \frac{x(0)^2}{2}$$

$$= \frac{x^3}{2}$$

Then,

$$\int_0^3 \int_0^{\sqrt{9-z^2}} \left[\int_0^x xydy\right] dxdz = \int_0^3 \int_0^{\sqrt{9-z^2}} \left(\frac{x^3}{3}\right) dxdz$$

$$= \int_0^3 \left[\int_0^{\sqrt{9-z^2}} \left(\frac{x^3}{2}\right) dx\right] dz$$

Integrate with respect to x, treat y as a constant

$$\int_0^{\sqrt{9-z^2}} \left(\frac{x^3}{2}\right) dx = \left(\frac{x^4}{8}\right)_0^{\sqrt{9-z^2}}$$

$$= \frac{\left(\sqrt{9-z^2}\right)^4}{8} - \frac{(0)^4}{8}$$

$$= \frac{1}{8}\left(9-z^2\right)^2$$

$$= \frac{1}{8}\left(81 - 18z^2 + z^4\right)$$

$$\int_0^3 \left[\int_0^{\sqrt{9-z^2}} \left(\frac{x^3}{3}\right) dx\right] dz = \frac{1}{8}\int_0^3 \left(81 - 18z^2 + z^4\right) dz$$

Integrate

$$= \frac{1}{8}\left(81z - 6z^3 + \frac{z^5}{5}\right)_0^3$$

Evaluate

$$= \frac{1}{8}\left(81\,(3) - 6(3)^3 + \frac{(3)^5}{5}\right) - \frac{1}{12}(0)$$

$$= \frac{1}{8}\left(\frac{648}{5}\right)$$

$$= \frac{81}{5}$$

Question 6:

$$\int_1^3 \int_x^{x^2} \int_0^{\ln z} x e^y \, dy \, dz \, dx$$

Solution:

$$\int_1^3 \int_x^{x^2} \int_0^{\ln z} x e^y dydzdx$$

$$= \int_1^3 \int_x^{x^2} \left[\int_0^{\ln z} x e^y dy \right] dzdx$$

Solve the inner integral, integrate with respect to y,
treat z and x as constants

$$\int_0^{\ln z} x e^y dy = [x e^y]_0^{\ln z}$$

Evaluate the limits in the variable y

$$= x e^{\ln z} - x e^0$$

$$= xz - z$$

Then,

$$\int_1^3 \int_x^{x^2} \left[\int_0^{\ln z} x e^y dy \right] dzdx = \int_1^3 \int_x^{x^2} (xz - x) \, dzdx$$

$$= \int_1^3 \left(\int_x^{x^2} (xz - x) \, dz \right) dx$$

Integrate with respect to z, treat x as a constant

$$\int_x^{x^2} (xz - x) \, dz = \left(\frac{xz^2}{2} - xz \right)_x^{x^2}$$

$$= \left(\frac{x \left(x^2\right)^2}{2} - x \left(x^2\right) \right) - \left(\frac{x (x)^2}{2} - x (x) \right)$$

$$= \frac{x^5}{2} - x^3 - \frac{x^3}{2} + x^2$$

$$= \frac{x^5}{2} - \frac{3x^3}{2} + x^2$$

$$\int_1^3 \left(\int_x^{x^2} (xz - x)\, dz \right) dx = \int_1^3 \left(\frac{x^5}{2} - \frac{3x^3}{2} + x^2 \right) dx$$

Integrate

$$= \left(\frac{x^6}{12} - \frac{3x^4}{8} + \frac{x^3}{3} \right)_1^3$$

Evaluate

$$= \left(\frac{(3)^6}{12} - \frac{3(3)^4}{8} + \frac{(3)^3}{3} \right) - \left(\frac{(1)^6}{12} - \frac{3(1)^4}{8} + \frac{(1)^3}{3} \right)$$

$$= \frac{315}{8} - \frac{1}{24}$$

$$= \frac{118}{3}$$

Question 7:

$$\int_0^2 \int_0^{\sqrt{4-x^2}} \int_{-5+x^2+y^2}^{3-x^2-y^2} x\, dz\, dy\, dx$$

Solution:

$$= \int_0^2 \int_0^{\sqrt{4-x^2}} \int_{-5+x^2+y^2}^{3-x^2-y^2} xdz\, dy\, dx$$

$$= \int_0^2 \int_0^{\sqrt{4-x^2}} [2x(4-x^2) - 2xy^2]\, dydx$$

$$= \int_0^2 \frac{4}{3} x(4-x^2)^{\frac{3}{2}}\, dx$$

$$= \frac{128}{15}$$

Question 8:

$$\int_1^2 \int_z^2 \int_0^{\sqrt{3}y} \frac{y}{x^2+y^2}\,dx\,dy\,dz$$

Solution:

$$\int_1^2 \int_z^2 \int_0^{\sqrt{3}y} \frac{y}{x^2+y^2}dxdydz$$

$$= \int_1^2 \int_z^2 \left[\int_0^{\sqrt{3}y} \frac{y}{x^2+y^2}dx\right]dydz$$

Solve the inner integral, integrate with respect to x, treat y and z as constants

$$\int_0^{\sqrt{3}y} \frac{y}{x^2+y^2}dx = \left[\arctan\left(\frac{x}{y}\right)\right]_0^{\sqrt{3}y}$$

Evaluate the limits in the variable x

$$= \arctan\left(\frac{\sqrt{3}y}{y}\right) - \arctan\left(\frac{0}{y}\right)$$

$$= \frac{\pi}{3} - 0$$

Then,

$$\int_1^2 \int_z^2 \left[\int_0^{\sqrt{3}y} \frac{y}{x^2+y^2}dx\right]dydz = \int_1^2 \int_z^2 \left(\frac{\pi}{3}\right)dydz$$

$$= \frac{\pi}{3}\int_1^2 \left(\int_z^2 dy\right)dz$$

Integrate with respect to y, treat z as a constant

$$\int_z^2 dy = (z)_z^2 = 2 - z$$

$$\frac{\pi}{3}\int_1^2 \left(\int_z^2 dy\right) dz = \frac{\pi}{3}\int_1^2 (2-z)dz$$

Integrate

$$= \frac{\pi}{3}\left(2z - \frac{z^2}{2}\right)_1^2$$

Evaluate

$$= \frac{\pi}{3}\left(2\,(2) - \frac{(2)^2}{2}\right) - \frac{\pi}{3}\left(2\,(1) - \frac{(1)^2}{2}\right)$$

$$= \frac{\pi}{3}(2) - \frac{\pi}{3}\left(\frac{3}{2}\right)$$

$$= \frac{1}{6}\pi$$